Statistical Mechanics of Money:

Applying Boltzmann-Gibbs Distributions and Entropy
to Economic Dynamics via Agent-Based Simulation

Submitted for:
Physics of Complex Systems (Exam)
Università della Magistrale in Fisica

September 5, 2026

Contents

Abstract

Keywords: Statistical mechanics, Boltzmann-Gibbs distribution, Gibbs entropy, Agent-based simulation, Markov processes, Economic dynamics, Complex systems.

1 Introduction

The applicability of **statistical mechanics** to economic systems, specifically how money distribution in a society of interacting economic agents follows the **Boltzmann-Gibbs distribution**. [1, 2] This is a foundational principle from thermodynamics and statistical physics. The assumption is that money, like energy in a physical system, is conserved during transactions between agents.

I conducted agent based simulations modeling economic transactions as random exchanges of money between agents, analogous to particle collisions in a gas. Under various transaction mechanisms such as constant exchanges between agents, fraction γ of individual wealth per agent in each transaction, fraction of system average, etc., the simulated money distribution converges to an exponential form $P(m) = \frac{1}{T}e^{-m/T}$, where $T = M/N$ represents the system's effective "temperature" (average wealth per agent).

Two simulation models are analyzed:

1. **Basic money distribution** with no debt constraints

2. **System with debt** allowing limited negative balances

It's possible to demonstrate that **Gibbs entropy** correctly predicts equilibrium states and that the system evolves as a **Markov process** toward maximum entropy. These results validate the hypothesis that economic inequality and wealth concentration emerge as natural equilibrium states when agents interact via random transactions, analogously to how physical systems spontaneously approach thermodynamic equilibrium.

This research connects fundamental concepts from statistical mechanics from Markov processes, Boltzmann-Gibbs distributions, Gibbs entropy, and dynamical systems theory to economics and the so called "econophysics" or "social physics", providing a computational validation of theoretical predictions.

1.1 Motivation

The term "econophysics" was introduced by analogy with similar terms, such as astrophysics, geophysics, and biophysics, which describe applications of physics to different fields.

It should be emphasized that econophysics does not literally apply the laws of physics, such as Newton's laws or quantum mechanics, to humans, but rather uses mathematical methods developed in statistical physics to study properties of complex economic systems consisting of a large number of humans. So, it may be considered as a branch of applied theory of probabilities. Econophysics distances itself from the verbose, narrative, and ideological style of political economy and is closer to econometrics in its focus. Studying mathematical models of a large number of interacting economic agents, econophysics has much common ground with the agent-based modeling and simulation [3].

1.2 Boltzmann-Gibbs Distribution of energy

The fundamental law of equilibrium statistical mechanics is the Boltzmann–Gibbs distribution. It states that the probability $P(\varepsilon)$ of finding a physical system or subsystem in a state with the energy ε is given by the exponential function

$$P(\varepsilon) = ce^{-\varepsilon/T}, \tag{1}$$

where T is the temperature, and c is a normalizing constant. Here we set the Boltzmann constant k_B to unity by choosing the energy units for measuring the physical temperature T. Then, the expectation value of any physical variable x can be obtained as

$$\langle x \rangle = \frac{\sum_k x_k e^{-\varepsilon_k/T}}{\sum_k e^{-\varepsilon_k/T}}, \tag{2}$$

where the sum is taken over all states of the system. Temperature is equal to the average energy per particle: $T \sim \langle \varepsilon \rangle$, up to a numerical coefficient of the order of 1.

Equation (1) can be derived in different way. One of the shortest derivations can be summarized as follows. Let us divide the system into two (generally unequal) parts. Then, the total energy is the sum of the parts: $\varepsilon = \varepsilon_1 + \varepsilon_2$, whereas the probability is the product of probabilities: $P(\varepsilon) = P(\varepsilon_1)P(\varepsilon_2)$. The only solution of these two equations is the exponential function (1).

A more sophisticated derivation, proposed by Boltzmann himself, uses the concept of **entropy**. Let us consider N particles with the total energy E. Let us divide the energy axis into small intervals (bins) of width $\Delta\varepsilon$ and count the number of particles N_k having the energies from ε_k to $\varepsilon_k + \Delta\varepsilon$. The ratio $N_k/N = P_k$ gives the probability for a particle to have the energy ε_k. Let us now calculate the **multiplicity** W, which is the number of permutations of the particles between different energy bins such that the occupation numbers of the bins do not change. This quantity is given by the combinatorial formula in terms of the factorials

$$W = \frac{N!}{N_1!N_2!N_3!\ldots}. \tag{3}$$

The logarithm of multiplicity is called the entropy $S = \ln W$. In the limit of large numbers, the entropy per particle can be written in the following form using the Stirling approximation for the factorials

$$\frac{S}{N} = -\sum_k \frac{N_k}{N} \ln\left(\frac{N_k}{N}\right) = -\sum_k P_k \ln P_k. \tag{4}$$

Now we would like to find what distribution of particles between different energy states has the highest entropy, i.e., the highest multiplicity, provided that the total energy of the system,

$$E = \sum_k N_k \varepsilon_k, \tag{5}$$

has a fixed value. Solution of this problem can be easily obtained using the method of Lagrange multipliers [?], and the answer gives the exponential distribution (1).

The same result can be derived from the **ergodic theory**, which says that the many-body system occupies all possible states of a given total energy with equal probabilities. Then it is straightforward to show [?] that the probability distribution of the energy of an individual particle is given by Eq. (1).

2 Model and Theory

As previously discussed, we can assume that agents can be modeled as particles in a closed system, where money is conserved. Instead of exchanging energy, agents exchange money through random transactions. The system evolves through random transactions, forming a Markov process. The theoretical framework predicts that the wealth distribution converges to the Boltzmann-Gibbs distribution, with entropy increasing over time until equilibrium is reached.

2.1 Conservation of Money

One could assume that Boltzmann's derivation of the energy distribution can be applied to other statistical systems with a conserved quantity, such as money. Indeed, we can consider the economy as a closed system with a fixed number of agents N and a total amount of money M. Therefore, the total money in the system is conserved. As a starting point, we can consider a simple model where agents exchange money through random transactions, and the total amount of money remains constant.

Introducing the concept of money conservation we can assume the model **without debt**, where agents cannot have negative balances. So the conditions for this model are:

- $\sum_{i=1}^{N} m_i = M$.
- $m_i \geq 0$, for $i = 1, \ldots, N$.
- $\delta m_i = -\delta m_j$, for $i \neq j$.
- $\delta_m < m_i$.

2.2 Boltzmann-Gibbs Distribution of Money

As argued in previous section, we can associate the money distribution to the energy exchange in a physical system. For this reason, we can assume that the probability of finding an agent with money

m is given by the Boltzmann-Gibbs distribution:

$$P(m) = ce^{-m/T}, \tag{6}$$

where c is a normalization constant and T is the effective temperature of the system: $T = M/N$, which is average money per agent.

The **entropy** of the system is defined as:

$$S = -\sum_{i=1}^{N} P(m_i) \log P(m_i). \tag{7}$$

The **equilibrium** is reached when the entropy is maximized when the system "cools off" to the Boltzmann-Gibbs distribution.

Many different kinds of transactions can be considered, in this simulation and report only two were used:

- **Constant transactions:** $\Delta m = 1$ where a fixed amount of money is exchanged between agents.
- **Fractional transactions** where a fraction γ of the agent's money is exchanged, the two agents will exchange only a portion of their money. This mechanism show how richer agents can exchange more money than poorer agents, and it is more realistic.

Both mechanisms showed the same behavior. They followed the Boltzmann-Gibbs distribution.

3 Results

3.1 Scenario 1: Basic Money Distribution

[Results content to be filled with figures and analysis...]

4 Conclusion

4.1 Summary

[Conclusion content to be filled...]

A Mathematical Details

A.1 Boltzmann-Gibbs Distribution

[Mathematical details to be filled...]

B Code Snippets

B.1 Core Simulation

[Code snippets to be filled...]

References

[1] Anirban Chakraborti and Bikas K. Chakrabarti. Statistical mechanics of money: how saving propensity affects its distribution. *The European Physical Journal B*, 17(1): 167–170, 2000.

[2] Adrian Dragulescu and Victor M Yakovenko. Statistical mechanics of money. *The European Physical Journal B-Condensed Matter and Complex Systems*, 17(4):723–729, 2000.

[3] Victor M. Yakovenko and J. Barkley Rosser. Colloquium: Statistical mechanics of money, wealth, and income. *Reviews of Modern Physics*, 81(4):1703–1725, 2009.